Laura E. Barnes, Ph.D.

Professor & Associate Chair, Systems & Information Engineering
Associate Director, Link Lab

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RESEARCH

My research advances the human-centered design of intelligent systems that integrate mobile sensing, wearable technologies, and data science to improve health and well-being. I study how human behaviors, captured across micro-level daily experiences and macro-level social and environmental contexts, shape health outcomes. Central to my work is translating these insights into clinical and community settings, where adaptive, AI-enabled systems can deliver personalized interventions and support proactive health management. By bridging ubiquitous computing, artificial intelligence, and human-centered design, my research produces both individualized and population-level insights that advance precision health and promote equitable, effective care.

APPOINTMENTS

School of Engineering and Applied Science, University of Virginia
Department of Systems & Information Engineering

- Associate Chair, 2025–present
- Professor (with Tenure), 2022–present
- Associate Director, Link Lab, 2023–present
- Director, Sensing Systems for Health (S^{2}He) Lab, 2012–present
- Associate Professor (with Tenure), 2018–2022
- Affiliated Faculty, *Data Science Institute* 2013–2016
- Full Member, *UVA Cancer Center* 2012–present
- Assistant Professor, 2012–2017

School of Internal Medicine and Division of Evidence-Based Medicine
University of South Florida

- Research Assistant Professor, 2010–2012

Automation and Robotics Research Institute, University of Texas Arlington

- Research Associate, 2008–2010

Department of Computer Science, University of South Florida

- Graduate Research Assistant, 2003–2008

Grid Computing Initiative, IBM Global Services

- Software Intern, 2001–2003

EDUCATION

- Ph.D., Computer Science, University of South Florida, 2008
- M.S., Computer Science, University of South Florida, 2007
- B.S., Computer Science & Mathematics, Texas Tech University, 2003

HONORS AND AWARDS

2024 UVA Research Achievement Award

2023 UVA Research Achievement Award

2021 Best Paper, ACM/IEEE Conference on Connected Health: Applications, Systems, and Engineering Technologies (CHASE)

2020 Recognition from Society of P.R.I. Award

2019 INFORMS Best Graduate Adviser Award

2018 Best Paper Award, ACM International Conference on Information System and Data Mining (ICISDM)

2017 Center for Design and Health Faculty Fellow

2015 Faculty Fellowship Summer Institute in Israel

2015 $25,000 Voto Latino Innovators Challenge, *Culturally-Competent Health IT for U.S. Latino Migrant Farmworker Population*

2015 Invited Speaker, Oak Ridge National Labs Biomedical Science and Engineering Conference

2015 Invited Keynote Speaker, NASA Ames Workshop on Transition to Autonomy

2015 Society of Behavioral Medicine Citation Award for Excellent Submission

2012 Excellence in Diversity (EDF) Fellowship, University of Virginia Teaching Resource Center

2005–2008 Oak Ridge Institute for Science and Education (ORISE) Graduate Fellowship

GRANTS, GIFTS, AND CONTRACTS

Under Review – PI, **National Science Foundation (NSF)**. *Upper EXTremity Examination for Neuromuscular Diseases (U-EXTEND): Evaluating Treatment Efficacy with Digital Multimodal Biomarkers of Muscle Structure and Function.* $1,200,000.

Under Review – MPI, **National Institutes of Health (NIMH)**. *Leveraging Personal Health Informatics to Build the Social Anxiety, Sensing and Social Interactions (SASSI) System.* $1,529,718.

Under Review – MPI, **National Institutes of Health (NIMH)**. *Harnessing Multimodal AI to Reduce Virologic Failure in People Living with HIV.* $5,512,436.

2026–2028 – Co-PI, **Thriving Youth in a Digital Environment (TYDE)**. *WellConnected: A Digital Tool to Help Young Adults Cultivate a Healthy Friendship Network.* $100,000.

2026–2027 – Co-PI, **Thriving Youth in a Digital Environment (TYDE)**. *GroundsWell: Discovering the Temporal Dynamics of Person–Environment Fit in College Students.* $30,000.

2025–2026 – PI, **National Science Foundation (NSF)**. *Student Travel Grant for the 2025 IEEE International Conference on Smart Computing (SmartComp).* $30,000.

2024–2025 – Co-PI, **Precision Health Initiative Community–University Partnership Grants**. *Addressing Racial Disparities in Suicide-Related Youth Outcomes Through Community Partnerships.* $25,000.

2024–2026 – PI, **National Institutes of Health (NIMH) R01MH132138-03S1**. *A Mixed-Methods Study of the Ethical Issues Surrounding Mobile Sensing in Digital Mental Health Interventions.* $151,434.

2023–2026 – Mentor, **Betty Irene Moore Fellowship for Nurse Leaders and Innovators (#9048)**. *Developing and Testing a Novel Mobile Health Solution (CommSense) to Improve Patient–Provider Communication.* $450,000.

2022–2025 – PI, **National Institutes of Health (Locus Health/NCI) SBIR Phase II Topic 406**. *Cancer Navigation Platform.* $200,000.

2022–2026 – PI, **National Institutes of Health (NIMH) R01MH132138-01**. *Context-Aware Micro-Interventions for Social Anxiety.* $1,200,000.

2022–2024 – Co-PI, **Coulter Translational Research Partnership**. *Upper EXTremity Examination for Neuromuscular Diseases (U-EXTEND): Objectively Assessing Treatment Efficacy and Accelerating Drug Discovery.* $231,847.

2022–2023 – Co-PI, **Engineering in Medicine**. *Exploring the Use of Wearable Sensors and Natural Language Processing Technology to Improve Provider–Patient Communication.* $86,000.

2021–2022 – Co-PI, **Engineering in Medicine**. *Wearable Technology–Guided Exercise Preconditioning to Accelerate Return of Function After Cancer Surgery.* $84,709.

2021–2023 – Co-PI, **UVA President's and Provost's Fund**. *Hoos Think Calmly.* $200,000.

2020–2023 – Co-PI, **Lung Cancer Foundation & Pfizer (LCRF-Pfizer #10478)**. *Real-Time Monitoring and Modeling of Symptoms and Adverse Events in Lung Cancer Patients Receiving Oral Targeted Therapies.* $350,000.

2020–2022 – Co-PI, **Engineering in Medicine**. *Upper EXTremity Examination for Neuromuscular Diseases (U-EXTEND): Technology-Enabled Methods to Assess Treatment Efficacy and Accelerate Drug Discovery.* $80,495.

2020–2021 – Co-PI, **Engineering in Medicine**. *Control of Light Exposure and Its Effect on Sleep Efficiency.* $87,000.

2020–2021 – Co-PI, **Global Infectious Diseases Institute Grant**. *Online Intervention to Reduce COVID-19 Anxiety and Predict Mental Health Trajectories Over Time.* $96,811.

2020–2021 – Co-PI, **Commonwealth Cyber Initiative (CCI)**. *Smart and Connected Health Research.* $50,000.

2019–2022 – PI, **National Institutes of Health (NIH) R01CA239246-01**. *Multiscale Modeling and Intervention for Improving Long-Term Medication Adherence in Context.* Total $1,200,000; UVA Share $435,131.

2018–2020 – Co-PI, **National Science Foundation (NSF) #1827674**. *PFI-RP: A Smart Building for*

Enhancing Human Performance, Comfort and Health. $337,432.

2018–2021 – PI, **Defense Advanced Research Projects Agency (DARPA) #GI15018**. *Warfighter Analytics Using Smartphones for Health (REaDI Sense).* $2,068,752.

2017–2021 – Co-I, **National Institutes of Mental Health (NIMH) R01MH113752-01**. *Effectiveness of Interpretation Training to Reduce Anxiety.* Total $2,039,218; LEB Share $284,843.

2016–2020 – Co-I, **National Institutes of Health (NIH) R01DK108957-01A1**. *Treating Type 2 Diabetes by Reducing Postprandial Glucose Elevations.* $2,726,735; LEB Share $100,000.

2015–2017 – PI, **Jeffress Trust Awards in Interdisciplinary Science (#151670-92634)**. *A New Computational Framework for Prediction of Severe Sepsis and Response to Therapy.* $100,000.

2015–2017 – PI, **National Institute of Aerospace / NASA #2A75**. *Real-Time Monitoring and Assessment of Pilot Performance and Cognitive State.* $250,000.

2015–2019 – PI, **Hobby Predoctoral and Postdoctoral Fellowships in Computational Science**. *Dynamic Monitoring of College Students' Mental Health Using Big Data Analytics.* $791,491.

2015–2017 – PI, **Coulter Translational Research Partnership**. *Superbug Tracker – Design of Context-Aware Surveillance System for Nosocomial Outbreaks.* $200,000.

2016–2017 – PI, **Coulter Translational Research Partnership**. *Optimized Predictive Monitoring of Non-Intensive Care Patients at Risk for Sepsis.* $100,000.

2016–2019 – Co-I, **National Science Foundation (NSF) #1560282**. *NSF REU Site: Multi-Scale Systems Bioengineering.* $361,861.

2015–2018 – Co-I, **National Science Foundation (NSF) #1461162**. *NSF REU Site: Wireless Health.* $299,077.

2015–2016 – Co-I, **UVA Cancer Center**. *Supporting Patients' Needs as Breast Cancer Survivors Through mHealth Assisted Self-Management.* $25,000.

2014–2015 – PI, **Ivy Biomedical Innovation Fund**. *Design of an Interactive Virtual Patient and Provider Platform for Interprofessional Teamwork Objective Structured Clinical Examinations.* $60,785.

2013–2017 – PI, **National Institutes of Health (NIH) R21CA167418**. *Feasibility of Virtual Agent Cervical Cancer Education for Hispanic Farmworkers.* Total $250,000; UVA Share $125,000.

2014–2017 – PI, **NIH Diversity Supplement to R21CA167418**. *Feasibility of Virtual Agent Cervical Cancer Education for Hispanic Farmworkers.* $23,697.

2016–2017 – Co-I, **Plastic Surgery Foundation (PSF)**. *Recovery from Mastectomy: Qualifying Survivorship Through Patient-Centered Outcomes.* $10,000.

2014–2015 – PI, **Virginia Space Grant Consortium**. *Data Fusion Techniques for Advanced Coordination-of-Care in the ICU.* $11,000.

2010–2012 – PI, **Patterson Foundation – Bring Science Home (#HSC-6001100502)**. *Design and Development of a Patient-Centric Diabetes Management System.* $100,000.

2010–2012 – Co-PI, **U.S. Army Medical Research and Materiel Command (#W81XWH-09-2-0175)**. *Development of EBM-CDSS to Aid Prognostication in Terminally Ill Patients.* $486,810.

PUBLICATIONS

Refereed Journal Articles

(†: Graduate Student or PostDoc Supervised by LEB; ‡: Undergraduate Student Supervised by LEB; ∗: LEB Supervisor or Co-Supervisor of Work; IF: Impact Factor; AR: Acceptance Rate)

93. †S. Kumar, †A. Rahman, †R. Gutierrez, S. Livermon, A. N. McCrady, S. Blemker, R. Scharf, L. E. Barnes. "A shape-based functional index for objective assessment of pediatric motor function." *PLoS One*, 20(10): e0332383, 2025. **[IF: 2.6]**

92. †Z. Wang, R. Yan, S. Francis, C. Diaz, T. Flickinger, Y. Lin, X. Hu, L. E. Barnes, V. LeBaron. "Stakeholder-centric participation in large language models enhanced health systems." *npj Health Systems*, 2:22, 2025, pp. 1-8.

91. E. R. Toner, †M. Rucker, †Z. Wang, M. A. Larrazabal, †L. Cai, †D. Datta, †H. Lone, M. Boukhechba, B. A. Teachman, L. E. Barnes. "Wearable sensor-based multimodal physiological responses of socially anxious individuals in social contexts on Zoom." *IEEE Transactions on Affective Computing*, 2025, pp. 1-12. **[IF: 9.8]**

90. C. C. Donahue, L. E. Barnes, J. N. Hertel, J. E. Resch. "Acute changes in sleep stages following concussion in collegiate athletes: A pilot study." *Journal of Athletic Training*, 2025. **[IF: 4.2]**

89. †N. Kaur, M. Gonzales IV, C. G. Alcaraz, J. Gong, K. J. Wells, L. E. Barnes. "A computational framework for longitudinal medication adherence prediction in breast cancer survivors: A social cognitive theory based approach." *PLOS Digital Health*, 4(6), 2025, pp. 1-21.

88. K. Petz, E. Toner, †M. Rucker, E. Leventhal, ∗S. Livermon, B. Davidson, M. Boukhechba, L. E. Barnes, B. Teachman. "Examining state affective and cognitive outcomes following brief mobile phone-based training sessions to reduce anxious interpretations." *Cognitive Therapy and Research*, 2025, pp. 1-23.

87. †A. Vela de la Garza, J. W. Eberle, †S. Baee, E. C. Wolf, M. Boukhechba, D. H. Funk, B. A. Teachman, L. E. Barnes. "Behavioral engagement patterns and psychosocial outcomes in web-based interpretation bias training for anxiety." *PLOS Digital Health*, 4(7), 2025, pp. 1-23.

86. ∗S. Livermon, A. M., Y. Zhang, K. Petz, E. Toner, †M. Rucker, M. Boukhechba, L. E. Barnes, B. A. Teachman. "A mobile intervention to reduce anxiety among university students, faculty, and staff: Mixed methods study on users' experiences." *PLOS Digital Health*, 4(1), 2025, pp. 1-26.

85. †Z. Wang, †F. Yuan, V. LeBaron, T. Flickinger, L. E. Barnes. "PALLM: Evaluating and Enhancing PALLiative Care Conversations with Large Language Models." *ACM Transactions on Computing for Healthcare*, 2025, pp. 1-24.

84. †Z. Wang, †N. Hassan, V. LeBaron, T. Flickinger, D. Ling, J. Edwards, C. Wu, M. Boukhechba, L. E. Barnes. "CommSense: A Wearable Sensing Computational Framework for Evaluating Patient-Clinician Interactions." *Proceedings of the ACM on Human-Computer Interaction*, 8(CSCW2), 2024, pp. 1-31. **[AR: 26%]**

83. A. N. McCrady, C. D. Masterson, L. E. Barnes, R. J. Scharf, S. S. Blemker. "Development of an ultrasound-based metric of muscle functional capacity for use in patients with neuromuscular disease." *Muscle and Nerve*, 70(6), 2024, pp. 1205-1214.

82. E. L. Leventhal, E. R. Toner, B. Davidson, M. Boukhechba, L. E. Barnes, B. A. Teachman. "Adapting a Mobile Anxiety Intervention for a University Community: Insights from a Qualitative Analysis." *Journal of Technology in Behavioral Science*, 2024, pp. 1-15.

81. I. Ladis, B. C. McGuinn, J. J. Glenn, A. L. Nobles, L. E. Barnes, B. A. Teachman. "Emerging Adults With a Suicide Attempt History Text More Emotionally With Peers than Family Members." *Emerging Adulthood*, 2023, pp. 1-12. **[IF: 2.6]**

80. †S. Baee, J. W. Eberle, †A. N. Baglione, T. Spears, E. Lewis, H. Wang, D. H. Funk, B. A. Teachman, L. E. Barnes. "Early Attrition Prediction for Web-Based Interpretation Bias Modification to Reduce Anxious Thinking: A Machine Learning Study." *JMIR Ment Health*, 11:e51567, 2024. **[IF: 4.8]**

79. I. Ladis, A. Seitov, L. E. Barnes, B. A. Teachman. "Perceived Burdensomeness and Thwarted Belongingness in Text Messages of Suicide Attempt Survivors." *Archives of Suicide Research*, 2023, pp. 1-12. **[IF: 2.8]**

78. J. W. Eberle, M. Boukhechba, J. Sun, D. Zhang, D. H. Funk, L. E. Barnes, B. A. Teachman. "Shifting Episodic Prediction With Online Cognitive Bias Modification: A Randomized Controlled Trial." *Clinical Psychological Science*, 11(5), 2023, pp. 819-840. **[IF: 2.465]**

77. V. LeBaron, T. Flickinger, D. Ling, H. Lee, J. Edwards, A. Tewari, Z. Wang, L. E. Barnes. "Feasibility and acceptability testing of CommSense: A novel communication technology to enhance health equity in clinician–patient interactions." *Digital Health*, 9, 2023.

76. †Z. Wang, M. A. Larrazabal, †M. Rucker, E. R. Toner, K. E. Daniel, †S. Kumar, M. Boukhechba, B. A. Teachman, L. E. Barnes. "Detecting Social Contexts from Mobile Sensing Indicators in Virtual Interactions with Socially Anxious Individuals." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, 7(3), 2023, pp. 1-26. **[IF: 4.163]**

75. M. A. Larrazabal, J. W. Eberle, †A. Vela de la Garza Evia, M. Boukhechba, D. H. Funk, L. E. Barnes, S. M. Boker, B. A. Teachman. "Online cognitive bias modification for interpretation to reduce anxious thinking during the COVID-19 pandemic." *Behaviour Research and Therapy*, 173, 2023, pp. 1-50. **[IF: 4.134]**

74. †G. Dong, †M. Tang, †Z. Wang, J. Gao, S. Guo, †L. Cai, †R. Gutierrez, B. Campbell, L. E. Barnes, M. Boukhechba. "Graph Neural Networks in IoT: A Survey." *ACM Transactions on Sensor Networks*, 19(2), 2023, pp. 1-50. **[IF: 4.1]**

73. †L. Cai, L. E. Barnes, M. Boukhechba. "Designing adaptive passive personal mobile sensing methods using reinforcement learning framework." *Journal of Ambient Intelligence and Humanized Computing*, 14, 2023, pp. 3019–3040. **[IF: 6.163]**

72. †N. J. Napoli, C. L. Stephens, K. D. Kennedy, L. E. Barnes, E. J. Garcia, A. R. Harrivel. "NAPS Fusion: A framework to overcome experimental data limitations to predict human performance and cognitive task outcomes." *Information Fusion*, 91, 2023, pp. 15-30. **[IF: 17.564]**

71. K. E. Daniel, M. A. Larrazabal, M. Boukhechba, L. E. Barnes, B. A. Teachman. "State and Trait Emotion Regulation Diversity in Social Anxiety." *Clinical Psychological Science*, 11(5), 2023, pp. 894-909. **[IF: 2.115]**

70. C. S. Kulkarni, S. Deng, T. Wang, J. Hartman-Kenzler, L. E. Barnes, S. Henrickson Parker, S. D. Safford, N. Lau. "Scene-dependent, feedforward eye gaze metrics can differentiate technical skill levels of trainees in laparoscopic surgery." *Surgical Endoscopy*, 37, 2023, pp. 1569-1580. **[IF: 3.149]**

69. I. Ladis, T. L. Valladares, D. L. Coppersmith, J. Glenn, †A. L. Nobles, L. E. Barnes, B. A. Teachman. "Inferring sleep disturbance from text messages of suicide attempt survivors: A pilot study." *Suicide and Life-Threatening Behavior*, 53(1), 2023, pp. 39-53. **[IF: 1.333]**

68. A. L. Silverman, J. M. Boggs, J. W. Eberle, M. Baldwin, H. C. Behan, A. Baglione, V. Paolino, A. Vela de la Garza, M. Boukhechba, L. E. Barnes, D. H. Funk, B. A. Teachman. "Minimal effect of messaging on engagement in a digital anxiety intervention." *Professional Psychology: Research and Practice*, 2023.

67. S. M. Rikard, B. Kim, J. D. Michel, S. M. Peirce, L. E. Barnes, M. D. Williams. "Identifying individual social risk factors using unstructured data in electronic health records and their relationship with adverse clinical outcomes." *SSM - Population Health*, 2022. **[IF: 4.7]**

66. †S. Kumar, †D. Datta, †G. Dong, †L. Cai, M. Boukhechba, L. Barnes. "Leveraging Mobile Sensing and Bayesian Change Point Analysis to Monitor Community-scale Behavioral Interventions: A Case Study on COVID-19." *ACM Transactions on Computing for Healthcare*, 3(4), 2022, pp. 1-13.

65. R. Gutierrez, A. McCrady, C. Masterson, S. Tolman, M. Boukhechba, L. Barnes, S. Blemker, R. Scharf. "Upper Extremity Examination for Neuromuscular Diseases (U-EXTEND): Protocol for a Multimodal Feasibility Study." *JMIR Research Protocols*, 11(10):e40856, 2022.

64. I. Ladis, E. R. Toner, A. R. Daros, K. E. Daniel, M. Boukhechba, P. I. Chow, L. E. Barnes, B. A. Teachman, B. Q. Ford. "Assessing Emotion Polyregulation in Daily Life: Who Uses It, When Is It Used, and How Effective Is It?" *Affective Science*, 2023. **[IF: 2.1]**

63. †Z. Wang, H. Xiong, J. Zhang, S. Yang, M. Boukhechba, D. Zhang, L. E. Barnes. "From Personalized Medicine to Population Health: A Survey of mHealth Sensing Techniques." *IEEE Internet of Things Journal*, 9(17), 2022, pp. 15413-15434. **[IF: 10.6]**

62. V. LeBaron, M. Boukhechba, J. Edwards, T. Flickinger, D. Ling, L. E. Barnes. "Exploring the Use of Wearable Sensors and Natural Language Processing Technology to Improve Patient-Clinician Communication: Protocol for a Feasibility Study." *JMIR Research Protocols*, 11(5):e37975, 2022.

61. †Z. Wang, H. Xiong, †M. Tang, M. Boukhechba, T. E. Flickinger, L. E. Barnes. "Mobile Sensing in the COVID-19 Era: A Review." *Health Data Science*, 2022, pp. 1-13.

60. I. Ladis, A. R. Daros, M. Boukhechba, K. E. Daniel, P. I. Chow, M. L. Beltzer, L. E. Barnes, B. A. Teachman. "When and Where Do People Regulate Their Emotions? Patterns of Emotion Regulation in Unselected and Socially Anxious Young Adults." *Journal of Social and Clinical Psychology*, 41(4), 2022, pp. 326-364. **[IF: 1.946]**

59. I. Ladis, T. Valladares, D. D. L. Coppersmith, J. J. Glenn, †A. L. Nobles, L. E. Barnes, B. A. Teachman. "Inferring sleep disturbance from text messages of suicide attempt survivors: A pilot study." *Suicide and Life-Threatening Behavior*, 2022, pp. 1-15. **[IF: 1.33]**

58. A. Werntz, A. L. Silverman, H. Behan, S. K. Patel, M. Beltzer, M. O. Boukhechba, L. E. Barnes, B. A. Teachman. "Lessons Learned: Providing Supportive Accountability in an Online Anxiety Intervention." *Behavior Therapy*, 2022. **[IF: 3.228]**

57. M. L. Beltzer, †M. K. Ameko, K. E. Daniel, A. R. Daros, M. Boukhechba, ∗L. E. Barnes, B. A. Teachman. "Building an emotion regulation recommender algorithm for socially anxious individuals using contextual bandits." *British Journal of Clinical Psychology*, 61, 2022, pp. 51-72. **[IF: 3.308]**

56. M. K. Mattos, C. A. Manning, M. Quigg, E. M. Davis, L. Barnes, A. Sollinger, M. Eckstein, L. Ritterband. "Feasibility and Preliminary Efficacy of an Internet-Delivered Intervention for Insomnia in Individuals with Mild Cognitive Impairment." *Journal of Alzheimer's Disease*, 84(4), 2021, pp. 1539-1550. **[IF: 4.472]**

55. K. E. Daniel, †S. Mendu, †A. Baglione, †L. Cai, B. A. Teachman, ∗L. E. Barnes, M. Boukhechba. "Cognitive bias modification for threat interpretations: using passive mobile sensing to detect intervention effects in daily life." *Anxiety, Stress & Coping*, 2021. **[IF: 2.887]**

54. †L. Cai, ∗L. E. Barnes, M. Boukhechba. "Designing adaptive passive personal mobile sensing methods using reinforcement learning framework." *Journal of Ambient Intelligence and Humanized Computing*, 2021. **[IF: 7.104]**

53. J. L. Ji, †S. Baee, D. Zhang, C. P. Calicho-Mamani, M. J. Meyer, D. Funk, S. Portnow, ∗L. E. Barnes, B. A. Teachman. "Multi-session online interpretation bias training for anxiety in a community sample." *Behaviour Research and Therapy*, 2021. **[IF: 4.5]**

52. K. E. Daniel, F. R. Goodman, M. L. Beltzer, A. R. Daros, M. Boukhechba, L. E. Barnes, B. A. Teachman. "Emotion malleability beliefs and emotion experience and regulation in the daily lives of people with high trait social anxiety." *Cognitive Therapy and Research*, 44, 2020, pp. 1186-1198. **[IF: 2.897]**

51. E. A. Gulleen, †M. Ameko, J. E. Ainsworth, L. E. Barnes, C. C. Moore. "Predictive models of fever, ICU transfer, and mortality in hospitalized patients with neutropenia." *Critical Care Explorations*, 2(12), 2020, pp. 1-10.

50. †H. Rashid, †S. Mendu, K. E. Daniel, M. L. Beltzer, B. A. Teachman, M. Boukhechba, L. E. Barnes. "Predicting subjective measures of social anxiety from sparsely collected mobile sensor data." *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4(3), 2020, pp. 1-24.

49. A. R. Daros, K. E. Daniel, M. L. Beltzer, M. Boukhechba, P. I. Chow, L. E. Barnes, B. A. Teachman. "Relationships between trait emotion dysregulation and emotional experiences in daily life: An experience sampling study." *Cognition and Emotion*, 34(4), 2020, pp. 743-755. **[IF: 2.563]**

48. †A. N. Baglione, J. Gong, M. Boukhechba, K. J. Wells, L. E. Barnes. "Leveraging mobile sensing to understand and develop intervention strategies to improve medication adherence." *IEEE Pervasive Computing*, 19(3), 2020, pp. 24-36. **[IF: 4.41]**

47. K. E. Daniel, A. R. Daros, M. L. Beltzer, M. Boukhechba, L. E. Barnes, B. A. Teachman. "How anxious are you right now? Using ecological momentary assessment to evaluate the effects of

cognitive bias modification for social threat interpretations." *Cognitive Therapy and Research*, 44, 2020, pp. 538-556. **[IF: 2.897]**

46. M. Boukhechba, †A. N. Baglione, L. E. Barnes. "Leveraging mobile sensing and machine learning for personalized mental health care." *Ergonomics in Design*, 28(4), 2020, pp. 18-23. **[IF: 3.74]**

45. †L. Cai, M. Boukhechba, M. S. Gerber, L. E. Barnes, S. L. Showalter, W. F. Cohn, P. I. Chow. "An integrated framework for using mobile sensing to understand response to mobile interventions among breast cancer patients." *Smart Health*, 15, 2020.

44. J. J. Glenn, †A. L. Nobles, L. E. Barnes, B. A. Teachman. "Can text messages identify suicide risk in real time? A within-subjects pilot examination of temporally sensitive markers of suicide risk." *Clinical Psychological Science*, 8(4), 2020, pp. 704-722. **[IF: 3.74]**

43. J. R. Gordon, L. A. Dial, S. D. Mills, J. H. Drizin, L. E. Barnes, V. L. Malcarne, E. Klonoff, A. Vaughn, K. J. Wells. "Celebrity disclosure prompts public discourse regarding appearance changes related to breast cancer care: A content analysis of internet comments." *Journal of Communication in Healthcare*, 13(1), 2020, pp. 64-71.

42. †N. Napoli, †M. Demas, C. L. Stephens, K. D. Kennedy, A. R. Harrivel, L. E. Barnes, A. T. Pope. "Activation complexity: A cognitive impairment tool for characterizing neuro-isolation." *Scientific Reports*, 10, 2020. **[IF: 5.2]**

41. †L. Cai, †M. Boukhechba, M. S. Gerber, L. E. Barnes, S. L. Showalter, W. F. Cohn, P. I. Chow. "An integrated framework for using mobile sensing to understand response to mobile interventions among breast cancer patients." *Smart Health*, 15, 2020.

40. D. J. Cox, J. M. Owens, L. E. Barnes, M. Moncrief, †M. Boukhechba, S. Buckman, T. Banton, B. Wotring. "A pilot study comparing newly licensed drivers with and without autism and experienced drivers in simulated and on-road driving." *Journal of Autism and Developmental Disorders*, 2020, pp. 1-11. **[IF: 3.341]**

39. J. Roe, L. E. Barnes, †N. J. Napoli, J. Thibodeaux. "The restorative health benefits of a tactical urban intervention: An urban waterfront study." *Frontiers in Built Environment*, 5, 2019, p. 71.

38. †K. Kowsari, K. M. Jafari, M. Heidarysafa, S. Mendu, L. Barnes, D. Brown. "Text classification algorithms: A survey." *Information*, 10(4), 2019, p. 150.

37. J. M. Thuman, H. McMahon, P. Chow, M. Gerber, K. Dassoulas, L. Barnes, C. A. Campbell. "The performance of patient-worn actigraphy devices to measure recovery after breast reconstruction." *Plastic and Reconstructive Surgery - Global Open*, 7(10), 2019, e2503.

36. A. R. Daros, K. E. Daniel, †M. Boukhechba, P. I. Chow, L. E. Barnes, B. A. Teachman. "Impact of social anxiety and social context on college students' emotion regulation strategy use: An experience sampling study." *Motivation and Emotion*, 43, 2019, pp. 844–855. **[IF: 1.459]**

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10. M. A. Fields, E. Haas, S. Hill, C. Stachowiak, L. E. Barnes. “Effective robot team control methodologies for battlefield applications.” In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2009, pp. 5862-5867. **[AR: 49%]**

9. R. Garcia, L. E. Barnes. “A multi-UAV simulator utilizing X-Plane.” In: *2nd International Symposium on Unmanned Aerial Vehicles*, 2009, pp. 1-8.

8. L. E. Barnes, M. A. Fields, K. Valavanis. “A bar magnet approach to controlling multirobot systems.” In: *17th Mediterranean Conference on Control and Automation (MED)*, 2009, pp. 416-421.

7. R. Garcia, M. A. Fields, L. E. Barnes. “Adaptable formations utilizing heterogeneous unmanned systems.” In: *SPIE Defense and Security Conference*, 2009, pp. 1-11.

6. L. E. Barnes, R. Garcia, M. A. Fields, K. Valavanis. “Swarm formation control utilizing ground and aerial unmanned systems.” In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2008, pp. 4205-4205. **[AR: 49%]**

5. L. E. Barnes, M. A. Fields, K. Valavanis. “Unmanned ground vehicle swarm formation control using potential fields.” In: *15th Mediterranean Conference on Control and Automation (MED)*, 2007, pp. 1-8.

4. L. E. Barnes, W. Alvis, M. A. Fields, K. Valavanis, W. Moreno. “Swarm formation control with potential fields formed by bivariate normal functions.” In: *14th Mediterranean Conference on Control and Automation (MED)*, 2006, pp. 1-7.

3. L. E. Barnes, R. R. Murphy, J. Craighead. “Human-robot coordination using scripts.” In: *SPIE Defense and Security Conference*, 2006, pp. 1-9.

2. L. E. Barnes, R. Garcia, T. Quasny, L. Pyeatt. “Multi-agent mapping using dynamic allocation utilizing a centralized storage system.” In: *12th Mediterranean Conference on Control and Automation (MED)*, 2004, pp. 1-6.

1. T. Fincannon, L. E. Barnes, R. R. Murphy, D. L. Riddle. “Evidence of the need for social intelligence in rescue robots.” In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2004, vol. 2, pp. 1089-1095. **[AR: 55%]**

Book Chapters

2. L. E. Barnes, R. Garcia, M. A. Fields, K. Valavanis. “Adaptive swarm formation control for hybrid ground and aerial assets.” In: *Cutting Edge Robotics 2010*, ed. V. Kordic. In-Tech, 2010, Chap. 23, pp. 263-278.

1. R. Garcia, L. E. Barnes, K. P. Valavanis. “Design of a hardware and software architecture for unmanned vehicles: A modular approach.” In: *Applications of Intelligent Control to Engineering Systems*, ed. K. P. Valavanis. Springer, 2009, pp. 91-115.

Refereed Abstracts and Posters (Select)

13. P. I. Chow, Y. Huang, J. Gong, M. Rucker, W. Bonelli, K. Leach, K. Fua, M. S. Gerber, M. Boukhechba, B. A. Teachman, L. E. Barnes. "Using smartphone sensors to understand behavioral dynamics of social anxiety." In: *9th Meeting of the International Society for Research on Internet Interventions*, 2017, Berlin, Germany.

12. J. R. Gordon, L. E. Barnes, G. Arroyo, P. Medina-Ramirez, S. Proctor, K. J. Wells. "Acceptability of a virtual patient educator tablet application to increase cervical cancer screening knowledge among Latinas." *Annals of Behavioral Medicine*, 51(Suppl 1):S770, 2017.

11. †J. V. Pulido, M. A. Fields, L. E. Barnes. "On-the-fly learning and monitoring of partially observed navigation plan." In: *AAMAS*, 2017, pp. 1700-1702.

10. L. A. Dial, J. R. Gordon, S. D. Mills, J. H. Drizin, L. Gutierrez, L. E. Barnes, V. Malcarne, E. Klonoff, A. Vaughn, K. J. Wells. "Elicitation of body image and appearances concerns following a celebrity's disclosure of prophylactic mastectomy." In: *American Psychosocial Oncology Society Annual Conference*, 2016.

9. E. Gulleen, Y. Wang, J. Ainsworth, L. E. Barnes, C. Moore. "Predictive models of fever and mortality in hospitalized patients with neutropenia." *Critical Care Medicine*, 2016.

8. C. Moore, J. Ainsworth, Y. Wang, L. E. Barnes. "Optimized predictive monitoring of non-intensive care unit patients at risk for sepsis." *Critical Care Medicine*, 2016.

7. ‡A. Chaet, ‡B. Morshedi, C. Brown, A. Mitchell, K. Wells, R. Valdez, L. E. Barnes. "Feasibility of employing health information technology to deliver health information to distinct Latino farmworker communities." In: *HFES Health Care Symposium*, 2015.

6. E. J. Charles, †N. J. Napoli, C. A. Foster, D. A. Good, T. B. Parker, E. A. Sharp, L. E. Barnes, J. S. Young. "Anticoagulant use contributes to increasing mortality after ground-level fall." In: *AAST Annual Meeting*, 2015.

5. †D. Datta, †W. Cosby, V. Brashers, J. Owen, C. White, L. E. Barnes. "Design of interactive patient and provider platform for interprofessional teamwork structured clinical examinations." In: *HFES Health Care Symposium*, 2015.

4. S. D. Mills, J. H. Drizin, E. A. Klonoff, L. E. Barnes, V. L. Malcarne, A. A. Vaughn, K. J. Wells. "Cognitive and behavioral change in response to a celebrity's health disclosure." In: *Society for Behavioral Medicine Annual Meeting*, 2015.

3. K. Saulters, A. Shankar, A. Majwala, R. Ssekitoleko, M. Auma, S. Adakun, A. Flower, L. E. Barnes, C. Moore. "A clinical prediction model of in-hospital mortality derived from patients admitted with acute infection to a regional referral hospital in Mbarara, Uganda." In: *American Society of Tropical Medicine and Hygiene*, 2015.

2. †J. Zhang, †H. Wu, J. Turner, A. Keller, L. E. Barnes. "Predicting future anxiety and depression diagnoses among college students utilizing electronic health data." In: *AMIA*, 2015.

1. ‡B. Morshedi, ‡A. Chaet, C. Brown, G. Arroyo, S. K. Proctor, R. S. Valdez, K. J. Wells, L. E. Barnes. "User preferences influencing the design of a tailored virtual patient educator in a Latina farm worker community." In: *AMIA*, 2014.

Conference Papers and Abstracts (Not Critically Reviewed)

29. B. Biber, M. Dodge, M. Gonzalez, R. Huang, O. Johnson, Z. Martin, A. Sieger, V. Tran, S. Xiao, L. E. Barnes. "Investigating the efficacy of virtual experiences on stress reduction." In: *IEEE Systems and Information Engineering Design Symposium (SIEDS)*, 2020, pp. 1-6.

28. C. Burley, D. Anderson, A. Brownlee, G. Lafer, T. Luong, M. McGowan, J. Nguyen, W. Trotter, H. Wine, A. Baglione, L. E. Barnes. "Increasing engagement in eHealth interventions using personalization and implementation intentions." In: *IEEE SIEDS*, 2020, pp. 1-5.

27. E. K. Barrett, C. M. Fard, H. N. Katinas, C. V. Moens, L. E. Perry, B. E. Ruddy, S. D. Shah, I. S. Tucker, T. J. Wilson, M. Rucker, L. Cai, L. E. Barnes, M. Boukhechba. "Mobile sensing: Leveraging machine learning for efficient human behavior modeling." In: *IEEE SIEDS*, 2020, pp. 1-7.

26. J. Azevedo, H. Delaney, M. Epperson, C. Jbeili, S. Jensen, C. McGrail, H. Weaver, A. Baglione, L. E. Barnes. "Gamification of eHealth interventions to increase user engagement and reduce attrition." In: *IEEE SIEDS*, 2019, pp. 1-5.

25. R. M. Karanth, M. S. Guyer, N. L. Twilley, M. B. Crosier, S. C. Monroe, A. J. McQuain, T. K. Lynn, M. Boukhechba, M. S. Gerber, L. E. Barnes. "Modeling user context from smartphone data for recognition of health status." In: *IEEE SIEDS*, 2019, pp. 1-5.

24. R. Stevens, K. Polk, C. Merrill, F. Feng, M. Weiss, E. Brosnan, M. S. Gerber, L. E. Barnes. "User experience design to enhance the effectiveness of mobile technologies for the treatment of mental health." In: *IEEE SIEDS*, 2018, pp. 1-6.

23. L. Schmeider, A. Kessenich, B. Libowitz, D. Hazlett, A. Pendharkar, B. Dickie, L. E. Barnes, M. S. Gerber. "Actigraphy-based modeling of patient recovery following breast reconstruction surgery." In: *IEEE SIEDS*, 2018, pp. 1-6.

22. B. Abraham, D. Duong, M. George, M. Greene, J. Kepley, K. Rosenburg, M. S. Gerber, L. E. Barnes. "Using patient-generated health data to facilitate preoperative decision making for breast cancer patients." In: *IEEE SIEDS*, 2017, pp. 1-6.

21. B. Alexander, K. Karakas, C. Kohout, H. Sakarya, N. Singh, J. Stachtiaris, L. E. Barnes, M. S. Gerber. "A behavioral sensing system that promotes positive lifestyle changes and improves metabolic control among adults with type 2 diabetes." In: *IEEE SIEDS*, 2017, pp. 1-6.

20. M. Buhl, J. Famulare, C. Glazier, J. Harris, A. McDowell, G. Waldrip, L. E. Barnes, M. S. Gerber. "Optimizing multi-channel health information delivery for behavioral change." In: *IEEE SIEDS*, 2016, pp. 130-135.

19. T. Bui, M. Grayson, K. Hofer, K. McGuire, M. Morrow, N. Rodammer, A. Farooque, L. E. Barnes, S. Patek. "Remote patient monitoring for improving outpatient care of patients at risk for sepsis." In: *IEEE SIEDS*, 2016, pp. 136-141.

18. I. Clark, A. Rigaut, A. Mathers, D. Brown, L. E. Barnes. "Survival analysis of carbapenemase-producing Enterobacteriaceae infections in hospital patients." In: *IEEE SIEDS*, 2016, pp. 95-100.
17. E. A. Eklund, O. M. Kassar, W. F. Barnhardt, N. J. Napoli, L. E. Barnes, J. S. Young. "An examination of expected and observed mortality in the setting of mono- and polytrauma." In: *Academic Surgical Congress*, 2016.
16. S. Mitchell, K. Schinkel, Y. Song, Y. Wang, J. Ainsworth, T. Halbert, S. Strong, J. Zhang, C. C. Moore, L. E. Barnes. "Optimization of sepsis risk assessment for ward patients." In: *IEEE SIEDS*, 2016, pp. 107-112.
15. M. Charles, T. N. Miano, X. Zhang, L. E. Barnes, J. M. Lobo. "Monitoring quality indicators for screening colonoscopies." In: *IEEE SIEDS*, 2015, pp. 171-175.
14. M. Craft, B. Dobrenz, E. Dornbush, M. Hunter, J. Morris, M. Stone, L. E. Barnes. "An assessment of visualization tools for patient monitoring and medical decision making." In: *IEEE SIEDS*, 2015, pp. 212-217.
13. J. Guillen, J. Liu, M. Furr, T. Wang, S. Strong, C. C. Moore, A. Flower, L. E. Barnes. "Predictive models for severe sepsis in adult ICU patients." In: *IEEE SIEDS*, 2015, pp. 182-187.
12. A. Harinder, L. Yesbeck, J. Farag, K. Mosquera, N. Napoli, W. Barnhardt, J. S. Young, J. Valdez, L. E. Barnes. "A design framework for evaluating changes in clinical communication systems." In: *IEEE SIEDS*, 2015, pp. 224-229.
11. O. M. Kassar, E. A. Eklund, W. F. Barnhardt, N. J. Napoli, L. E. Barnes, J. S. Young. "Trauma survival margin analysis: A dissection of trauma center performance through initial lactate." In: *Southeastern Surgical Congress*, 2015.
10. J. Stern, S. Hewitt, M. Guilfoyle, C. Mishra, A. Mathers, J. Lobo, D. Brown, L. E. Barnes. "Modeling nosocomial transmission of carbapenem-resistant bacteria." In: *IEEE SIEDS*, 2015, pp. 176-181.
9. L. E. Barnes, G. Arroyo, S. S. Proctor, C. Brown, A. Chaet, B. Morshedi, K. J. Wells. "Design of mobile, virtual patient educator for delivering cancer education to Latina farmworkers." In: *22nd Annual Conference on Women's Health*, 2014.
8. L. E. Barnes, K. J. Wells. "Design of interactive health information technology for low health literacy users." In: *Stanford MedX Conference*, 2014.
7. N. Napoli, D. Lake, R. Moorman, L. E. Barnes. "Fusion of electrocardiogram leads to improve detection of cardiac waveforms in an intensive care setting." In: *International Conference on Complexity in Acute Illness*, 2014.
6. L.-V. Ngo, K. Walker, A. Holowinsky, G. Knox, R. Shimberg, N. Napoli, J. Gillen, J. Young, L. E. Barnes. "Analysis of communication patterns in critical care environments." In: *IEEE SIEDS*, 2014, pp. 129-134.
5. C. Pearce, M. Guckenberg, B. Holden, A. Leach, R. Hughes, C. Xie, M. Hassett, A. Adderley, L. E. Barnes, M. Sherri, G. C. Lewin. "Designing a spatially aware, automated quadcopter using an Android control system." In: *IEEE SIEDS*, 2014, pp. 23-28.

4. Y. Wu, D. Samant, K. Squibbs, A. Chaet, B. Morshedi, L. E. Barnes. "Design of interactive cancer education technology for Latina farmworkers." In: *IEEE SIEDS*, 2014, pp. 1-4.
3. M. P. Mackrell, K. J. Twilley, W. P. Kirk, L. Q. Lu, J. L. Underhill, L. E. Barnes. "Discovering anomalous patterns in network traffic data during crisis events." In: *IEEE SIEDS*, 2013, pp. 52-57.
2. L. E. Barnes, M. Rivera, C. D. Meade, S. K. Proctor, L. M. Gutierrez, K. J. Wells. "Feasibility study for technology-based cancer education for Latinas from an agricultural community." In: *Cancer Epidemiology, Biomarkers Prevention (supplement)*, 2011.
1. T. Fincannon, L. Barnes, R. Murphy, D. Riddle. "Talking and gesturing to a robot: Emergent social interaction in rescue robots." In: *2nd Annual USF Undergraduate Research Symposium*, 2004.

Tutorials

1. M. Rucker, S. Ahmed, L. E. Barnes. "Deploying smart sensors for human subjects." IEEE SmartComp Research Tutorial, June 2025.

Panels

2. R. Valdez, S. Myneni, A. Hartzler, L. Mamykina, L. E. Barnes. "Opportunities for social media within consumer health informatics." In: *American Medical Informatics Association Symposium*, 2015.
1. E. Montague, R. Valdez, M. Burns, L. E. Barnes, M. Vaughn-Cooke. "Considering culture in the design of health information technology: Lessons learned and experiences." In: *Human Factors and Ergonomics Society Annual Meeting*, 2014.

Patents, Disclosures, and Copyrights

4. Systems and methods for context-aware anxiety interventions, 2024.
3. Upper EXTremity Examination for Neuromuscular Diseases (U-EXTEND): Objectively Assessing Treatment Efficacy and Accelerating Drug Discovery, 2022.
2. Optimization and prediction of sepsis and response to therapy in non-ICU patients (disclosure), 2017.
1. Superbug Tracker – Design of context-aware surveillance system for nosocomial outbreaks involving non-patient reservoirs (disclosure), 2015.

TEACHING

University of Virginia (Instructor)

∗ Developed new or significantly revised curriculum † Co-developed course taught by postdoc

- ∗ **Computational Models for AI Systems** (SYS 2201) [Undergraduate] — Fall 2023, Fall 2024, Fall 2025. Enrollment: 20–65
- ∗ **Statistical Modeling I** (SYS 6021) [AMP Graduate] — Spring 2023, Fall 2024, Fall 2025. Enrollment: 10–20
- ∗ **Machine Learning** (CS 6316 / SYS 6015) [Graduate] — Spring 2013, Spring 2014. Enrollment: 20–35
- ∗ **Programming for Information Engineering** (SYS 3501) [Undergraduate] — Fall 2023, Fall 2024. Enrollment: 20–46
- ∗ **Foundations of Statistical Modeling** (SYS 4021 / SYS 6021) [Undergraduate/Graduate] — Fall 2012–2022. Enrollment: 80–150
- ∗ **Cooperative Autonomous Systems** (ECE 6502 / SYS 6582) [Graduate] — Spring 2017. Enrollment: 20
- † **Mobile Sensing and Health** (SYS 4581 / SYS 6581 / PSYCH 4559 / PSYCH 7681) [Undergraduate/Graduate] — Fall 2016–2019. Enrollment: 10–25
- ∗ **AMP Statistics for Engineers** (APMA 6430) [Graduate] — Fall 2019–2021. Enrollment: 15–30
- ∗ **AMP Agent-Based Modeling and Simulation** (SYS 7002) [Graduate] — Spring 2013–2016. Enrollment: 18–40
- ∗ **AMP Machine Learning** (SYS 7002) [Graduate] — Spring 2017–2019. Enrollment: 18–30
- **Senior Design Capstone I and II** (SYS 4053 / SYS 4054) [Undergraduate] — 2012–2025. Enrollment: 5–20
- **Data Science Capstone** (DSI 6011) [Undergraduate] — 2014–2016. Enrollment: 3–10

University of South Florida (Instructor)

- ∗ **Evidence-Based Clinical Reasoning Module** (BMS 6387) [Graduate] — 2010–2012. Enrollment: 80–100
- ∗ **Software Engineering** (CEN 4020) [Undergraduate] — Summer 2007. Enrollment: 30

University of Denver (Instructor)

- ∗ **Software Engineering for Mechatronic Systems** (COMP 3704 / COMP 3381) [Graduate] — 2009–2010. Enrollment: 35

INVITED TALKS

- **Invited Speaker**, Emory University - School of Nursing, October 2025.
- **Invited Speaker**, IEEE Biomedical and Health Informatics Conference — Workshop on Digital Health Technologies and AI for Studying and Predicting Episodic Health Events, October 2025.
- **Invited Speaker**, IEEE Biomedical and Health Informatics Conference — Special Session on Intelligent and Timely Health Support Outside the Clinic, October 2025.
- **Invited Speaker**, South China Normal University — Aberdeen Institute of Data Science and Artificial Intelligence, June 2025.
- **Featured Speaker**, National Science Foundation — Smart Health Frontiers: Combating Cancer with Advanced Technologies, October 2024.
- **Invited Speaker**, Charlottesville Women in Tech, September 2024.
- **Invited Speaker**, School of Data Science Datapalooza, November 2023.
- **Keynote Speaker**, AI in Health Conference: AI, Data, and Computing for Global Impact, Rice University, November 2022.
- **Speaker**, Society for Digital Mental Health Meeting, 2023.
- **Keynote Panelist**, COVID-19 Panel, IEEE/ACM Conference on Connected Health Applications, Systems, and Engineering Technologies, December 2021.
- **Leidos Lunch and Learn Speaker**, "Wearables for Smart Health," November 2021.
- **Invited Speaker**, Applied Science and Engineering and Department of Psychology — PsychEng Seminar Series, University of Toronto, November 2021.
- **Invited Speaker**, Department of Industrial and Systems Engineering, Oklahoma State University & University of Oklahoma, Norman, OK, April 2021.
- **Invited Speaker**, School of Engineering, Arizona State University, Tempe, AZ, March 2020.
- **Invited Speaker**, Department of Computer Science, University of Texas at Dallas, Richardson, TX, February 2020.
- **Panelist**, Tom Tom Founders Day Festival, Charlottesville, VA, April 2018.
- **Invited Speaker**, Department of Industrial and Systems Engineering, Virginia Tech, Blacksburg, VA, March 2017.
- **Invited Speaker**, Department of Information Systems, University of Maryland–Baltimore County, Baltimore, MD, April 2016.
- **Invited Speaker**, Department of Health Administration and Public Policy, George Mason University, Fairfax, VA, March 2016.
- **Invited Speaker**, Department of Public Health Sciences Grand Rounds, University of Virginia, March 2016.

- **Invited Speaker**, Department of Computer Science Seminar, Virginia Commonwealth University, Richmond, VA, March 2016.
- **Invited Speaker**, Department of Electrical and Computer Engineering Seminar, Virginia Commonwealth University, Richmond, VA, February 2016.
- **Invited Speaker**, Quantitative Psychology Research Group, University of Virginia, Charlottesville, VA, February 2016.
- **Speaker**, UVA Datapalooza, Charlottesville, VA, September 2015.
- **Speaker**, Annual Oak Ridge National Laboratory Biomedical Science and Engineering Conference, Oak Ridge, TN, August 2015.
- **Keynote Speaker**, Workshop on Transition to Autonomy, NASA Ames Research Center, Moffett Field, CA, March 2015.
- **Organizer & Speaker**, Workshop on Data Science Methods for Consequence Management of Cyber-Physical Systems, Charlottesville, VA, July 2014.
- **Invited Speaker**, Broadband Wireless Access & Applications Center, Auburn, AL, October 2013.
- **Invited Speaker**, Broadband Wireless Access & Applications Center Meeting, Tucson, AZ, April 2013.
- **Invited Speaker**, University of Virginia Cancer Center Seminar, Charlottesville, VA, October 2013.
- **Invited Speaker**, Wireless Health Seminar, Charlottesville, VA, February 2013.
- **Invited Speaker**, Biomedical Engineering Department Seminar, University of Virginia, Charlottesville, VA, January 2013.
- **Invited Speaker**, Computer Science & Engineering Department Seminar, University of Colorado–Colorado Springs, March 2012.
- **Invited Speaker**, Systems and Information Engineering Department Seminar, University of Virginia, Charlottesville, VA, February 2012.
- **Invited Speaker**, Electrical Engineering Department Seminar, Texas Tech University, Lubbock, TX, January 2011.
- **Invited Speaker**, College of Engineering Seminar, University of Texas Arlington, Fort Worth, TX, September 2007.
- **Invited Speaker**, Electrical and Computer Engineering Department & Lockheed Martin Seminar, University of Denver, Denver, CO, March 2009.
- **Invited Speaker**, Electrical & Computer Engineering Department, University of Denver, Denver, CO, August 2009.

INTERNAL SERVICE

Departmental Service

- Search Committee Member: SIE Department Chair, 2025–2026
- Search Committee Member: SIE T3 Faculty in OR and PHI Grand Challenges, 2024–2025
- Associate Chair, SIE, 2024–present
- Associate Chair for Research, 2021–2022
- Graduate Committee Chair/Co-Chair, 2019–2024
- Search Committee Member: ESE T3 Faculty Hire, 2021–2022
- Search Committee Chair: Modeling and Simulation at the Human–Technology Frontier, 2017–2019
- Search Committee Member: SIE Research Scientist, 2017–2018
- Search Committee Member: Accelerated Master's Program Director, 2016–2017
- Search Committee Member: Cloud-Scale Analytics T3 Cluster Hire, 2015–2016
- Search Committee Member: Data Science Instructor, 2014–2015
- Search Committee Member: Human Factors T3 Position, 2013–2014
- Organizer, Graduate Recruiting Day, 2014–2018
- Faculty Marshal, SIE Commencement Exercises, 2013–2018
- SIE Graduate Committee, 2013–present
- Undergraduate Advising, 2013–present

School Level Service

- SEAS Faculty Hiring Lead for PHI Grand Challenges, 2024–present
- SEAS Grant Brewing Initiative, 2024
- Commonwealth Cyber Initiative (CCI) Leadership Committee, 2023–present
- Associate Director, Link Lab, 2023–present
- Link Lab Leadership Committee, 2019–2020
- Graduate Office Working Group, 2016–2018
- Co-development of data science courses for digital health (with postdocs in engineering and psychology), 2015–2018
- Developed new graduate curriculum in robotics (Education Innovation Award), 2016–2017
- SEAS Open House Participant, 2014–2017

University Level Service

- Gun Violence Solutions Committee, 2025–present
- CIRCL (Contemplative Innovation + Research Co-Lab) Advisory Committee, 2024–present
- School of Data Science Promotion & Tenure Committee, 2024–present
- Faculty Director, Grand Challenge Digital Technology Core, 2023–present
- Internal Submission Reviewer, 2023–present
- Search Committee Member: School of Data Science T3 Position, 2019–2022
- School of Data Science Undergraduate Committee, 2019–present
- iTHRIV Faculty Mentorship Committees, 2018–2023
- Internal Review Committee, Jeffress Trust Awards, 2018–2019
- Informatics Director, College Health Surveillance Network (CHSN), 2016–2018
- Member, CITOC Innovations Subcommittee, 2014–present
- ASPIRE Steering Committee, 2014–present

EXTERNAL SERVICE

Reviewer for Journals and Conferences (Selected)

- IEEE/ACM CHASE
- IEEE Journal of Biomedical and Health Informatics
- IEEE Transactions on Big Data
- Journal of Medical Internet Research (JMIR)
- ACM UbiComp / ISWC
- BMC Medical Informatics & Decision Making
- IEEE Transactions on Biomedical Engineering
- IEEE Transactions on Systems, Man, and Cybernetics B
- IEEE Transactions on Robotics
- AMIA Annual Symposium
- IEEE ICRA, IROS, MED, EMBC

Peer Reviewer for Government Agencies

- Review Panelist, National Institutes of Health, 2014–2025
- Review Panelist, National Science Foundation, 2014–2024
- Reviewer, Swiss National Science Foundation, 2021, 2025
- Reviewer, Israel Science Foundation, 2018
- Panelist, ASEE Air Force Faculty Fellowship Program, 2016
- Panelist, National Defense Science & Engineering Graduate Fellowship Program, 2016
- Panelist, NSF Data Science Workshop, 2015
- Panelist, ASEE SMART Scholarship Program, 2014–2015

Committee Activities

- Editor, *Proc. ACM Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2025–present
- Co-General Chair, IEEE SmartComp, 2025
- Chair of Posters and Demos, ACM UbiComp / ISWC, 2025
- Associate Editor, *Proc. ACM IMWUT*, 2023–2025
- Associate Editor, *Nature Scientific Reports*, 2020–2023
- Associate Editor, *ACM Transactions on Health*, 2020–present
- Program Committee Member, ACM UbiComp (2016–2018), AIME (2017–2018), CHASE (2016–2019), Human Factors in Healthcare (2015–2017)
- Organizer & Presenter, Workshop on Quantitative Analysis and Data Mining for Clinicians, Int. Conf. on Complexity in Acute Illness, 2014
- Session Chair, IEEE EMBC, 2011

Leadership, Citizenship, and Mentoring

- Technical Lead, American College Health Association Data Warehouse Initiative, 2016–2018
- Mentor, VA-NC Alliance for Minority Participation, 2015–2017
- Adviser, Undergraduate Student Opportunities in Academic Research (USOAR), 2015–2017
- Founder, UVA *Sensing Systems for Health (S^2He)* Lab
- Speaker or Panelist: Women in Tech, STEM Forums, March Med-ness, SWE Majors Fair (2014–2016)
- Presenter, Holton Arms School & IBM GREAT Camp outreach, 2007–2014

ADVISING AND MENTORING

Postdoctoral Fellows and Research Scientists

- **Zachary King** — April 2026–present.
- **Indrajeet Ghosh** — November 2025–present.
- **Lihua Cai** — 2020–2021.
  Current: Assistant Professor, Aberdeen Institute of Data Science and Artificial Intelligence, South China Normal University; Chief Technology Officer, Prime Digital Technologies.
- **Haroon Rashid Lone** — 2018–2020.
  Current: Assistant Professor, Electrical Engineering and Computer Science, Indian Institute of Science Education and Research Bhopal.
- **Mehdi Boukhechba** — 2017–2019.
  Current: Senior Research Scientist, Johnson & Johnson.
- **Jiaqi Gong** — 2016–2017.
  Current: Associate Professor, Computer Science, University of Alabama.
- **Haoyi Xiong** — 2015–2016.
  Current: Principal Applied Scientist, Microsoft.

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PhD Students Advised

- **Alejandro Vistorte Salgado** — PhD Systems Engineering, 2025–present (expected 2029).
- **Xinyu Chen** — PhD Systems Engineering, 2025–present (expected 2029).
- **Sabbir Ahmed** — PhD Systems Engineering, 2023–present (expected 2028).
- **Ariful Islam** — PhD Systems Engineering, 2024–present (expected Fall 2026).
- **Arafat Rahman** — PhD Systems Engineering, 2022–present (expected Fall 2026).
- **Zhiyuan Wang** — PhD Systems Engineering, 2021–present (expected May 2025).
- **Mark Rucker** — PhD Systems Engineering, Fall 2025.
  Dissertation: Online Learning Framework for Predicting Social Context and Receptivity in Just-in-Time Adaptive Interventions for Social Anxiety Disorder.
  Current: Director, Digital Technology Core in Research Computing, University of Virginia.
- **Navreet Kaur** — PhD Systems Engineering, Spring 2024.
  Dissertation: Towards Improving Medication Adherence Leveraging Personalized, Theory-Informed Models.
  Current: Data Scientist, Citi Bank.
- **Shashwat Kumar** — PhD Systems Engineering, Summer 2024.
  Dissertation: Shape Analysis Methods for Motor Function Assessment.
  Current: Postdoctoral Research Associate, Johns Hopkins University.

- **Anna Baglione** — PhD Systems Engineering, Spring 2023.
  Dissertation: Modeling User Behavior in Context: A Systems-Level Approach to Mobile Health.
  Current: Data Scientist, Stanson Health.
- **Sonia Baee** — PhD Systems Engineering, Summer 2022.
  Dissertation: Computational Models of Engagement in Digital Health Platforms.
  Current: Machine Learning Engineer, Apple.
- **Mawulolo Ameko** — PhD Systems Engineering, Fall 2021.
  Dissertation: Computational Methods for Personalizing Mobile Health Interventions.
  Current: Data and Applied Scientist, Microsoft.
- **Lihua Cai** — PhD Systems Engineering, Spring 2020.
  Dissertation: Adaptive Mobile Sensing Using Reinforcement Learning Framework.
  Current: Assistant Professor, Aberdeen Institute of Data Science and Artificial Intelligence, South China Normal University; CTO, Prime Digital Technologies.
- **Kamran Kowsari** — PhD Systems Engineering, Spring 2020 (Co-advised with Donald Brown).
  Dissertation: Diagnosis and Analysis of Celiac Disease and Environmental Enteropathy on Biopsy Images Using Deep Learning Approaches.
  Current: Data Scientist, School of Medicine, University of California–Los Angeles.
- **Alicia Nobles** — PhD Systems Engineering, Fall 2018.
  Dissertation: Mining User-Generated Data to Examine Online Information Seeking for Reproductive and Sexual Health.
  Current: Principal Data Scientist, Walmart Health Data Science.
- **Nicholas Napoli** — PhD Systems Engineering, Spring 2018.
  Dissertation: Characterizing Uncertainty in Sensor Fusion to Improve Predictive Models.
  Current: Associate Professor, Electrical and Computer Engineering, University of Florida.
- **Jinghe Zhang** — PhD Systems Engineering, Fall 2017.
  Dissertation: Representation Learning from Longitudinal Electronic Health Record Data for Patient Characterization and Prediction of Health Outcomes.
  Current: Machine Learning Engineer, Apple.

MS Students Advised

- **Angel Vela de la Garza** — MS Systems Engineering, Summer 2022.
  Thesis: Understanding the Relationship Between Engagement Markers and Psychosocial Outcomes in a Digital Mental Health Intervention for Anxiety.
- **Sanjana Mendu** — MS Systems Engineering, Summer 2020.
  Thesis: SocialText: A Framework for Understanding Mental Health From Digital Communication Patterns.
  Current: Data Scientist, University of Virginia Health System.
- **Michael Clemens** — MS Systems Engineering, Summer 2020.
  Thesis: Perceived Trust of United States Politicians Through Prosody.
  Current: PhD Student, Computer Science, University of Utah.

- **Shabnam Wahed** — MS Computer Engineering, Fall 2019.
  Thesis: Dempster–Shafer Theory-Based Sensor Fusion for Context Recognition.
- **Ketki Vilankar** — MS Systems Engineering, Summer 2016.
  Thesis: Spatio-Temporal Visualization and Modeling of Nosocomial Infections.
- **Debajyoti Datta** — MS Systems Engineering, Spring 2016.
  Thesis: A Deep Learning Methodology for Semantic Utterance Classification in Domain-Specific Dialogue Systems.
- **Hao Wu** — MS Systems Engineering, Summer 2015.
  Thesis: A Methodology for Analyzing Healthcare Utilization Among College Students With Mental Health Disorders.
- **Matthew Demas** (co-advised with Nathan Lau) — MS Systems Engineering, Summer 2015.
  Thesis: Quantitative Modeling of Human Performance in Nuclear Power Plant Control Room Validation.
- **Callie Kwiatkowski** — MS Systems Engineering, Summer 2014.
  Thesis: Measuring Care Variation in Congestive Heart Failure Patients.

PhD Committees

- **Mohammadreza Ganjiarjenaki** — PhD Systems Engineering (in progress).
- **Elaheh Mohammadi** — PhD Systems Engineering (in progress).
- **Beatrice Li** — PhD Systems Engineering (in progress).
- **Wendy Qi** — PhD Systems Engineering (in progress).
- **Yeonbin Son** — PhD Systems Engineering (in progress).
- **Haley Green** — PhD Computer Engineering, Summer 2025.
  Dissertation: Modeling Trust in Human–Robot Teams.
- **Sihang Jiang** — PhD Systems Engineering, Summer 2025.
  Dissertation: Bayesian Risk Analysis for COVID-19.
- **Samarth Singh** — PhD Systems Engineering, Summer 2025.
  Dissertation: Sustainable Re-Design of Hydropower Systems for Renewable Energy Transitions and Climate Change Adaptation.
- **Jeremy Eberle** — PhD Psychology, Summer 2024.
  Dissertation: Measurement of Target Engagement and Network Analysis of Change Mechanisms in Web-Based Interpretation Bias Training for Anxiety.
- **Hossain Kaviani** — PhD Systems Engineering, Summer 2024.
  Dissertation: Designing Robust Control Rules for Stochastic Engineered Systems.
- **Alexandra Silverman** — PhD Psychology, Spring 2024.
  Dissertation: Examining Strategies for Increasing Engagement With Digital Mental Health Interventions Among Anxious Individuals.

- **Saurav Sengupta** — PhD Systems Engineering, Summer 2024.
  Dissertation: Combining and Evaluating Computer Vision and Natural Language Processing Techniques for Use in Clinical Contexts.
- **Lauren Bramblett** — PhD Systems Engineering, Fall 2024.
  Dissertation: Foundations of Epistemic Planning for Collaborative Multi-Robot Systems.
- **Tahsin Mullick** — PhD Systems Engineering, Fall 2024.
  Dissertation: Modeling Human Behavior With Passive Sensing: Modeling Paradigms, Limited Data, and Spatio-Temporal Patterns.
- **Katharine Daniel** — PhD Psychology, Summer 2023.
  Dissertation: Problems With Switching: Investigating the Sequence of Emotion Regulation Strategy Choices in the Daily Lives of Socially Anxious People.
- **Catherine Donahue** — PhD Kinesiology, Spring 2023.
  Dissertation: Characterizing Sleep-Related Changes and Their Effect on Recovery Following Concussion in Collegiate Athletes.
- **Runze Yan** — PhD Systems Engineering, Spring 2023.
  Dissertation: A Computational Framework for Modeling Cyclic Human Behaviors From Multi-Modal Sensor Data.
- **Mojtaba Heidarysafa** — PhD Systems Engineering, Spring 2023.
  Dissertation: Knowledge Discovery and Decision Support Systems Using Natural Language Processing in Applications for Societal Good.
- **Md Fazlay Rabbi Masum Billah** — PhD Computer Science, Fall 2023.
  Dissertation: Enhancing the Human–IoT Interaction Experience Through Deep Multimodal Sensor Fusion.
- **Payden McBee** — PhD Systems Engineering, Summer 2023.
  Dissertation: Image Segmentation in Histopathology With Limited Labeled Data.
- **Mofijul Islam** — PhD Systems Engineering, Summer 2023.
  Dissertation: Multimodal and Multitask Representation Learning for Perceiving Embodied Interactions.
- **Erfan Pakdamanian** — PhD Systems Engineering, Summer 2022.
  Dissertation: Predicting and Improving Takeover Performance by a Context-Aware Deep Learning–Based Assistive System.
- **Maria Phillips** — PhD Systems Engineering, Summer 2022.
  Dissertation: Moving the Dialogue Forward: Virtual Student Conversational Agent Design in Low Data Environments.
- **Robert Gutierrez** — PhD Systems Engineering, Summer 2022.
  Dissertation: Leveraging Ubiquitous Sensing for Quantifying the Quality of Motion in Mobile Health.
- **Guimin Dong** — PhD Systems Engineering, Spring 2022.
  Dissertation: Deep Graph Learning in Mobile Health.

- **Paul Bonczeck** — PhD Electrical Engineering, Fall 2022.
  Dissertation: Randomness-Based Behavior Monitoring for Resilient Autonomous Robot Operations.
- **Yichen Jiang** — PhD Systems Engineering, Fall 2022.
  Dissertation: Simulation and Modeling of Information Dissemination in Online Social Networks.
- **Sodiq Adewole** — PhD Systems Engineering, Fall 2021.
  Dissertation: Long Video Content Analysis: Learning to Summarize Capsule Endoscopy Video.
- **Miranda Beltzer** — PhD Psychology, Fall 2021.
  Dissertation: Examining Social Reinforcement Learning Biases in Social Anxiety.
- **Peng Wang** — PhD Electrical Engineering, Fall 2021.
  Dissertation: Ultra–Low Power Multi-Modal Sensor Interface Circuits and Systems for Personalized Physiological Monitoring.
- **William Adorno** — PhD Systems Engineering, Summer 2021.
  Dissertation: Hyperparameter Optimization of Deep Learning Models for Biomedical Image Segmentation.
- **Michaela Rikard** — PhD Biomedical Engineering, Summer 2021.
  Dissertation: Multiscale Modeling Applications in Cardiovascular Disease and Public Health.
- **Meltem Yucel** — PhD Psychology, Spring 2021.
  Dissertation: “No Fair!": An Investigation of Children’s Development of Fairness.
- **Vince Pulido** — PhD Systems Engineering, Spring 2021.
  Dissertation: Building Computer-Aided Diagnostic Models for Biopsy Images Using Minimally Curated Datasets.
- **Rasoul Sali** — PhD Systems Engineering, Spring 2021.
  Dissertation: Deep Representation Learning of Whole-Slide Histopathology Images.
- **Ankur Sarker** — PhD Computer Science, Spring 2021.
  Dissertation: Data-Driven Misbehavior Generation and Detection Methodologies for Intelligent Transportation Systems.
- **Nasrin Sadeghzadehyazdi** — PhD Computer Engineering, Fall 2020.
  Dissertation: A Skeleton-Based Study of Gait With Applications in LiDAR-Based Gait Recognition and Pathological Gait Identification.
- **Ridwan Alam** — PhD Computer Engineering, Fall 2020.
  Dissertation: Wearable Health Monitoring: Robust Modeling of Physiology and Behavior From Real-World Sparse-Labeled Sensing.
- **Krista Rand** — PhD Systems Engineering, Fall 2020.
  Dissertation: A System-Theoretic Approach to Dynamic Recovery Strategy Design in Networked Systems.
- **Sarah Preum** — PhD Computer Science, Summer 2020.
  Dissertation: Information Extraction and Fusion for Improving Health Safety.

- **Robert Moulder** — PhD Psychology, Spring 2020.
  Dissertation: Latent Multivariate Maximal Lyapunov Exponents.
- **Abu Mondol** — PhD Computer Science, Spring 2020.
  Dissertation: Toward Robust and Accurate Human Activity Recognition Using Wearable Sensors.
- **Jonathan Hughes** — PhD Systems Engineering, Fall 2019.
  Dissertation: Online Adaptive Personalization of Supervised Learning Models in Mobile Health Treatment Systems.
- **Mohamad Alipour** — PhD Civil Engineering, Fall 2019.
  Dissertation: Deep Learning for Robust and Efficient Automated Defect Recognition in Critical Infrastructure.
- **Adi Shaked** — PhD Psychology, Summer 2019.
  Dissertation: On the Same Wavelength: Shared Emotions as Information for Social Evaluation.
- **Basak Ozaslan** — PhD Systems Engineering, Summer 2019.
  Dissertation: Personalized Optimization of Insulin Treatment Policies for the Management of Physical Activity in Patients With Type 1 Diabetes.
- **Mohsin Ahmed** — PhD Computer Science, Summer 2019.
  Dissertation: Real-Time Dynamic Physical Phenomenon–Adapted Machine Learning Pipeline for Smart Acoustics.
- **Peter Wu** — PhD Systems Engineering, Spring 2019.
  Dissertation: Mining Social Signals in Cyber–Human Systems: Collective Behavior, Personal Health, and Modeling Methods.
- **Ifat Emi** — PhD Computer Science, Spring 2019.
  Dissertation: A System for Recognizing Complete and Partial Daily In-Home Activities and Monitoring Activity Quality.
- **Tamal Batabyal** — PhD Computer Engineering, Spring 2019.
  Dissertation: Graph-Theoretic Data Modeling With Application to Neuromorphology, Activity Recognition, and Event Detection.
- **Yang Wang** — PhD Systems Engineering, Summer 2018.
  Dissertation: Multi-Party Privacy-Preserving Machine Learning and Its Applications.
- **Jeffrey Glenn** — PhD Psychology, May 2018.
  Dissertation: Can Personal Electronic Communications Identify Suicide Risk in Real Time?
- **Seongah Hong** — PhD Civil Engineering, Fall 2017.
  Dissertation: Development of a Traffic Control Framework for Connected and Automated Vehicles Using Optimal Control Theory.
- **Xiaomin Lin** — PhD Systems Engineering, Fall 2017.
  Dissertation: Machine Learning Approaches to Multi-Agent Inverse Learning Problems.
- **Rituparna Sarkar** — PhD Computer Engineering, Spring 2017.
  Dissertation: Dictionary Learning and Sparse Representation for Image Analysis With Application to Segmentation, Image Retrieval, and Unusual Event Detection From Video.

- **Matthew Engelhard** — PhD Systems Engineering, Fall 2016.
  Dissertation: Adaptive Algorithms for Personalized Health Monitoring.
- **Kevin Leach** — PhD Computer Engineering, Fall 2016.
  Dissertation: Transparent System Introspection in Support of Analyzing Stealthy Malware.
- **Juhi Ranjan** — PhD Computer Science, Fall 2016.
  Dissertation: It Takes Two: Exploring Interactions Between Smart Objects and Wearables to Implicitly Identify and Authenticate Object Users.
- **Michael Landau** — PhD Systems Engineering, Summer 2016.
  Dissertation: Optimal 6D Object Pose Estimation With Commodity Depth Sensors.
- **Johnny Gan** — PhD Systems Engineering, Spring 2016.
  Dissertation: Applying Genetic Algorithms to the Problem of Variable Selection in Large Datasets With Interaction Terms.
- **Benjamin Brahim** — PhD Systems Engineering, Spring 2015.
  Dissertation: Enabling Physical Activity for Type 1 Diabetes Mellitus by Real-Time Risk Assessment and Insulin Dose Adjustment Advice.
- **Peng Su** — PhD Civil Engineering, Summer 2014.
  Dissertation: Highway Reservation System: Models, Simulations, and Implementation Discussions.
- **Junrui Xu** — PhD Systems Engineering, Spring 2014.
  Dissertation: Heterogeneous Data Uncertainties in Risk Management of a Multiscale Transportation Program.
- **Eleazar Gil-Herrera** — PhD Industrial and Management Systems Engineering, University of South Florida, Fall 2013.
  Dissertation: Classification Models in Clinical Decision Making.

Undergraduate Theses / Capstones Mentored

- **2019–20** — B. Biber, M. Dodge, M. Gonzalez, R. Huang, O. Johnson, Z. Martin, A. Sieger, V. Tran, S. Xiao.
  Investigating the Efficacy of Virtual Experiences on Stress Reduction.
- **2019–20** — C. Burley, D. Anderson, A. Brownlee, G. Lafer, T. Luong, M. McGowan, J. Nguyen, W. Trotter, H. Wine.
  Increasing Engagement in eHealth Interventions Using Personalization and Implementation Intentions.
- **2019–20** — E. K. Barrett, C. M. Fard, H. N. Katinas, C. V. Moens, L. E. Perry, B. E. Ruddy, S. D. Shah, I. S. Tucker, T. J. Wilson.
  Mobile Sensing: Leveraging Machine Learning for Efficient Human Behavior Modeling.
- **2019–20** — J. Azevedo, H. Delaney, M. Epperson, C. Jbeili, S. Jensen, C. McGrail, H. Weaver.
  Gamification of eHealth Interventions to Increase User Engagement and Reduce Attrition.
- **2018–19** — R. M. Karanth, M. S. Guyer, N. L. Twilley, M. B. Crosier, S. C. Monroe, A. J. McQuain, L. T. Kha.
  Modeling User Context From Smartphone Data for Recognition of Health Status.
- **2018–19** — J. de Paiva Azevedo, H. Delaney, M. Epperson, C. Jbeili, S. Jensen, C. McGrail, H. Weaver.
  Gamification of eHealth Interventions to Increase User Engagement and Reduce Attrition.
- **2017–18** — E. Brosnan, F. Feng, C. Merrill, K. Polk, R. Stevens, M. Weiss.
  User Experience Design to Enhance the Effectiveness of Mobile Technologies for Treatment of Mental Health.
- **2017–18** — B. Dickie, D. Hazlett, A. Kessenich, B. Libowitz, A. Penharkar, L. Schmeider.
  Using Patient-Generated Health Data to Facilitate Pre-Operative Decision Making in Breast Cancer Patients.
- **2016–17** — R. Hazard.
  Derivation and Validation of a Universal Vital Assessment (UVA) Score: A Tool for Predicting Mortality in Adult Hospitalised Patients in Sub-Saharan Africa.
- **2016–17** — B. Abraham, D. Duong, M. George, M. Greene, J. Kepley, K. Rosenburg.
  Using Patient-Generated Health Data to Facilitate Pre-Operative Decision Making in Breast Cancer Patients.
- **2016–17** — B. Alexander, K. Karakas, C. Kohout, H. Sakarya, N. Singh, J. Stachtiaris.
  Optimizing Smartphone-Based Content Delivery Strategies for Behavioral Change.
- **2015–16** — M. Buhl, J. Famulare, C. Glazier, J. Harris, A. McDowell, G. Waldrip.
  IRIS: Optimizing Multi-Channel Health Information Delivery for Behavioral Change.
- **2015–16** — N. Hogan, C. Lewenhaupt, E. Love, J. Smith.
  A System for the Tracking and Quantification of Depression in the College Population.
- **2015–16** — M. McCall.
  PET: Personalized Exercise Therapy "High" Assistive Technology for Children With Autism Spectrum Disorders.

- **2015–16** — A. Chaet.
  Designing Consumer Health Information Technology for the U.S. Latino Population: A Review of the Field and Original Research.
- **2015–16** — A. Mitchell.
  Educational Software for Low-Literacy Migrant Farmworkers.
- **2014–15** — E. Dornbush, B. Dobrenz, J. Morris, M. Craft, M. Hunter.
  Real-Time Processing and Visualization of Complex Physiologic Data for Patient Monitoring and Medical Decision Making.
- **2014–15** — A. Harinder, J. Farag, K. Mosquera, L. Yesbeck.
  Analysis of Communication Processes for Optimal Triage of ICU Trauma Patients.
- **2013–14** — D. Samant, K. Squibbs, Y. Wu.
  Design of Interactive Cancer Education Technology for Latina Farmworkers.
- **2013–14** — A. Howlinsky, G. Knox, L.-V. Ngo, C. Shimberg, K. Walker.
  Analysis of Communication Patterns in Critical Care Environments.
- **2012–13** — W. Kirk, M. Mackrell, K. Twilley, J. Underhill.
  Using Network Load and Traffic Indicators to Build Models of the Size and Severity of Disaster Events.

Selected Undergraduate Research Assistants

**Funded through REUs, Grants, USOAR, and the VA–NC Alliance.*

- **Neha Chopra** — Computer Science, Summer 2018–Fall 2019.
- **Courtney Jacobs** — Computer Science, Fall 2017–Summer 2020.
- **Sarai Alvarez** — Bioengineering, Summer 2017.
- **Philip Schroeder** — Human Biology, Spring 2017–Spring 2018.
- **Rohan Ravel** — Computer Science, Fall 2016–Spring 2017.
- **Roman Wang** — Computer Science, Fall 2016–Fall 2018.
- **Matthew Ledwith** — Systems and Information Engineering, Fall 2016–Spring 2017.
- **Catherine Sun** — Biostatistics, Fall 2016–present.
- **Olivia LeBeau** — Human Biology, Fall 2016–Spring 2017.
- **Heather Lukas** — Biomedical Engineering, Summer 2016.
- **Sanjana Medu** — Computer Science, Spring 2015–2018.
- **Wes Bonelli** — Computer Science, Summer 2015–present.
- **Riley Hazard** — Human Biology & Statistics, Summer 2015–2017.
- **Raven Morris** — Computer Science, Summer 2015.
- **Torian Wright** — Systems & Information Engineering, Fall 2015–Spring 2016.
- **Bijan Morshedi** — Chemistry, Fall 2013–Spring 2016.
- **Alexis Chaet** — Human Biology and Public Health, Fall 2013–Spring 2016.
- **Andrew Mitchell** — Computer Science, Spring 2014–Spring 2016.
- **Joy Ruiz** — Systems & Information Engineering, Summer 2013–Fall 2013.
- **William Cosby** — Physics & Computer Science, Spring 2013–Summer 2014.
- **Samuel Prescott** — Computer Science, Spring 2014–Summer 2014.
- **Michelle Stone** — Systems & Information Engineering, Fall 2013–Spring 2014.
- **Chetan Mishra** — Systems & Information Engineering, Spring 2015–Summer 2015.

SELECTED STUDENT AWARDS

- **2025** — Sabbir Ahmed, UVA Center for Global Inquiry and Innovation Award.
- **2025** — Zhiyuan Wang, Link Lab Distinguished Research Award.
- **2025** — Zhiyuan Wang, All-University Graduate Teaching Award.
- **2024** — Zhiyuan Wang, NSF Cyber-Physical Systems (CPS) Rising Stars.
- **2024** — Zhiyuan Wang, UVA Sture G. Olsson Endowed Fellowship.
- **2018** — Kamran Kowsari, Presidential Fellowship in Data Science Award for *Women and Cyber Recruitment of Extremist Groups: A Text Analysis Approach*.
- **2017** — Jinghe Zhang, IBM Social Good Fellowship (1 of only six awarded).
- **2016** — Alicia Nobles, NIH Biomedical Data Sciences Training Grant Appointment.
- **2016** — Alicia Nobles, NIH Big Data to Knowledge (BD2K) Fellowship: *Big Data Coursework for Computational Medicine (BDC4CM)*.
- **2016** — Yu Huang, Presidential Fellowship in Data Science Award for *Loneliness in an Age of Unprecedented Social Connection: Using Data Analytics to Explore the Role of Social Media in Bolstering or Impairing Well-Being*.
- **2016** — Nicholas Napoli, Metropolitan Washington Chapter of Achievement Rewards for College Scientists (MWC/ARCS) Foundation Scholar.
- **2016** — Alicia Nobles, Double Hoo Research Grant for *Health and Insurance Literacy Among College Students*.
- **2016** — Riley Hazard, Harrison Undergraduate Research Award for *Early Warning Scores for In-Hospital Mortality in Sub-Saharan Africa*.
- **2015** — Riley Hazard, Undergraduate Research Network Workshop & Conference Grant for Uganda Rural Risk Score Development.
- **2015** — Joseph Guillen, Jiankun Liu, Margaret Furr, Tianyao Wang, Best Paper Award (Health and Safety) at the IEEE Systems and Information Engineering Design Symposium for *Predictive Models for Severe Sepsis in Adult ICU Patients* (Data Science Institute Capstone).

SELECTED PRESS AND MEDIA

- **September 2025** — Work the Digital Technology Core is Enabling: UVA Study Hopes to Heal Confidence Along with Torn Ligaments.
- **March 2023** — Prof. Virginia LeBaron Earns Moore Fellowship to Develop Conversation Assessment Tool for Clinicians.
- **March 2023** — A Cognitive Assistant on Your Doctor's Smartwatch?.
- **January 2023** — Watch Your Words.
- **September 2022** — A Review on Mobile Sensing in the COVID-19 Era.
- **February 2020** — Does My Smartphone Know I Am Sick?.
- **January 2020** — New Research Finds Text Messages Can Help Predict Suicide.
- **January 2019** — Sensing Systems for Health Lab.
- **April 2018** — The Future of Patient Care.
- **February 2018** — Smart and Connected Health for Precision Medicine.
- **October 2017** — An End to Dieting for People with Type 2 Diabetes?.
- **September 2017** — Triage Tool Helps Doctors Save Lives by Identifying Patients at Greatest Risk.
- **July 2017** — Successful STEM Program, Led by UVA, Nets New Grant to Expand.
- **September 2016** — Female Engineers Discuss Ways to Prosper.
- **July 2015** — UVA Partners with U.S. Army Research Laboratory.